

Sharp pointwise bounds for Lebesgue functions

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Abstract

For each n , let λ_n be the Lebesgue function for polynomial interpolation at an arbitrary set of n distinct nodes in $[-1, 1]$. We prove that the points $x \in (-1, 1)$ for which $\lambda_n(x) > (2/\pi) \log n - C(x)$ infinitely often, with some finite $C(x)$, form a dense set. We also prove $\limsup_{n \rightarrow \infty} \lambda_n(x) / \log n \geq 2/\pi$ for almost every $x \in (-1, 1)$. We also show that no absolute additive constant works for all arrays: for every $M > 0$, there is an array for which each fixed $x \in (-1, 1)$ eventually satisfies $\lambda_n(x) \leq (2/\pi) \log n - M$. For the same array, no single finite constant gives a dense set of points with infinitely many lower-bound occurrences. The positive results use derivative-jump energies and positive Cauchy transforms. The counterexample combines an explicit logarithmic integral estimate on the gaps of a compact set with real-rooted polynomials having prescribed amplitudes. The proofs are formalized in Lean 4; generative AI was used in developing the arguments.

1 Introduction

For each $n \geq 1$, let

$$-1 \leq x_{1,n} < \cdots < x_{n,n} \leq 1.$$

Define the node polynomial, the cardinal polynomials, and the Lebesgue function by

$$P_n(x) = \prod_{k=1}^n (x - x_{k,n}),$$
$$\ell_{k,n}(x) = \frac{P_n(x)}{(x - x_{k,n})P'_n(x_{k,n})}, \quad \lambda_n(x) = \sum_{k=1}^n |\ell_{k,n}(x)|.$$

The cardinal polynomials are interpreted by continuity at the nodes. The rows of the array need not be nested.

Theorem 1. *For every triangular array of distinct nodes as above, the following statements hold.*

(i) *The set of points $x \in (-1, 1)$ for which there is a finite constant $C(x)$ such that*

$$\lambda_n(x) > \frac{2}{\pi} \log n - C(x) \tag{1.1}$$

for infinitely many n is dense in $(-1, 1)$.

(ii) *For Lebesgue almost every $x \in (-1, 1)$,*

$$\limsup_{n \rightarrow \infty} \frac{\lambda_n(x)}{\log n} \geq \frac{2}{\pi}. \tag{1.2}$$

Theorem 2. *For every $M > 0$, there is a triangular array of distinct nodes in $(-1, 1)$ such that, for every fixed $x \in (-1, 1)$,*

$$\lambda_n(x) \leq \frac{2}{\pi} \log n - M \quad \text{for all sufficiently large } n. \quad (1.3)$$

The starting index may depend on x . The array can also be chosen so that, for every finite C , the set

$$G_C = \left\{ x \in (-1, 1) : \lambda_n(x) > \frac{2}{\pi} \log n - C \text{ for infinitely many } n \right\} \quad (1.4)$$

is not dense in $(-1, 1)$.

Erdős [4, p. 68] asked whether every triangular array has a point at which the sharp logarithmic lower bound holds infinitely often with a bounded additive loss, and whether the normalized lower bound (1.2) holds almost everywhere. Theorem 1(i) answers the first question with a constant that may depend on the point; part (ii) answers the second. As Tao observes in [10, Remark 1.12], the original question does not specify the allowed dependence of the additive constant. Theorem 2 rules out two natural uniform strengthenings of Theorem 1(i).

Write $G_C(X)$ for the set in (1.4) associated with the array X . Consider the following statements, where $C \in \mathbb{R}$, X ranges over triangular arrays, and density is in $(-1, 1)$:

- (U1) $\exists C \forall X, \quad G_C(X) \neq \emptyset;$
- (U2) $\forall X \exists C, \quad G_C(X) \text{ is dense};$
- (U3) $\forall X, \quad \bigcup_{C \in \mathbb{R}} G_C(X) \text{ is dense.}$

Theorem 2 disproves (U1) and (U2), while Theorem 1(i) proves (U3).

Theorem 2 concerns arbitrary triangular arrays, as in [4, p. 68] and [10, Corollary 1.11]. The problem page [2] instead uses the initial segments of a single infinite node sequence. Our counterexample does not impose this nesting, and the uniform-constant question in that setting remains open.

1.1 Background

Faber [9] proved the order $\log n$ for the global Lebesgue constant, Bernstein [1] obtained the asymptotically sharp leading constant $2/\pi$, Erdős and Turán [6] improved the error to $O(\log \log n)$, Erdős [5] to $O(1)$, and Vértési [11, Corollary 1] refined the asymptotics through the optimal constant term to an error of $O((\log \log n / \log n)^2)$. See [10, Section 1.3] and [3] for further historical discussion.

For triangular arrays, Erdős [4, p. 68] stated that the set of points x satisfying

$$\limsup_{n \rightarrow \infty} \frac{\lambda_n(x)}{\log n} \geq \frac{2}{\pi}$$

is everywhere dense, and asked whether this holds for almost every x . Erdős and Vértési [7] proved almost-everywhere divergence of interpolants of a suitable continuous function for every array. Their subsequent paper [8, Theorem 2.1] gives a quantitative lower bound for the Lebesgue functions: for every $\varepsilon > 0$, there are $\eta(\varepsilon) > 0$ and $n_0(\varepsilon)$ such that, for every array and $n \geq n_0(\varepsilon)$, there is a measurable set $H_n \subset [-1, 1]$ satisfying

$$|H_n| \leq \varepsilon, \quad \lambda_n(x) > \eta(\varepsilon) \log n \quad (x \in [-1, 1] \setminus H_n).$$

They note that Chebyshev nodes show this order to be optimal. The exceptional sets may vary with n ; the estimate implies $\limsup_n \lambda_n(x)/\log n > 0$ almost everywhere (indeed, $|\bigcup_N \bigcap_{n \geq N} H_n| \leq \varepsilon$, so outside a set of measure at most ε , each point leaves H_n infinitely often; now let $\varepsilon \downarrow 0$). Thus the logarithmic order was already known in 1981. Theorem 1(ii) supplies the optimal coefficient $2/\pi$, which is the coefficient in Erdős's question and in [10, Remark 1.12]. Erdős and Vértési also give the integral version [8, Corollary 2.2]:

$$\int_{S_n} \lambda_n(x) dx > (|S_n| - \varepsilon) \eta(\varepsilon) \log n \quad (n \geq n_0(\varepsilon))$$

for arbitrary measurable $S_n \subset [-1, 1]$.

Corollary 11 gives, in every sufficiently large row, a point in each fixed positive-measure set with $\lambda_n(x) > c \log n$, for any $0 < c < 2/\pi$. Erdős–Vértési's theorem instead controls all points outside a small exceptional set, with coefficient $\eta(\varepsilon)$.

Tao [10, Theorem 1.10(i)] proved that every fixed nondegenerate interval $I \subset [-1, 1]$ satisfies

$$\sup_{x \in I} \lambda_n(x) \geq \frac{2}{\pi} \log n - O_I(1).$$

His Corollary 1.11 and Remark 1.12 give, for every prescribed $\omega(n) \rightarrow \infty$, a dense, indeed comeager, set of points x for which

$$\lambda_n(x) \geq \frac{2}{\pi} \log n - \omega(n)$$

infinitely often. Theorem 1(i) replaces $\omega(n)$ by a finite constant depending on the point, on a dense set. Corollary 12 shows that, for arbitrary triangular arrays, the additive constant in the interval lower bound cannot in general be chosen independently of the interval; see also [10, Remark 1.12].

1.2 Organization

Sections 2–4 prove Theorem 1(i). The local differentiation estimate turns large derivative jumps into nearby high values of the Lebesgue function. A weighted energy bound supplies enough such jumps, and a second-moment argument gives recurrence on a positive-measure set. Baire's theorem then gives density.

Sections 5–6 prove Theorem 1(ii) by a separate argument using positive Cauchy transforms. This proof is independent of Sections 2–4 and of Tao's estimates. Section 7 proves Theorem 2 independently of both lower-bound arguments. Its first three subsections construct positive functions with large logarithmic integral ratios; Lemma 10 converts them into interpolation rows, which are assembled in the final subsection. Section 8 gives the positive-measure and interval-constant corollaries.

Throughout the paper, all logarithms are natural and $|E|$ denotes Lebesgue measure. We write $c_0 = 2/\pi$, and suppress the row index when discussing a single row. Constants in local estimates may depend on the fixed intervals and test functions, but not on n or on interpolation data bounded by one. The notation $J \Subset U$ means that J is a compact subset of the open set U .

2 A local differentiation formula

We use the following local form of Riesz's differentiation formula and its near-equality consequence.

Lemma 3. Suppose that $T \geq 2\pi$ and F is holomorphic in a neighborhood of the disk $\{|z| \leq T\}$, is real on its real diameter, and satisfies, throughout this disk,

$$|F(u + iv)| \leq e^{|v|}.$$

Then

$$|F'(0)| \leq 1 + O(T^{-1}). \quad (2.1)$$

If $F'(0) \geq 1 - \delta$, where $\delta \geq 0$, then

$$F(\pi/2) \geq 1 - \frac{\pi^2}{4}\delta - O(T^{-1}). \quad (2.2)$$

The constants are absolute.

Proof. Choose $R = m\pi$ with $T/2 \leq R < T$, and integrate

$$\frac{F(z)}{z^2 \cos z}$$

around $|z| = R$. On this circle, $e^{|\Im z|} \leq 8|\cos z|$. Indeed, for $|v| \geq 1$, use $|\cos(u + iv)| \geq \sinh |v| \geq e^{|v|}/4$. For $|v| \leq 1$, put $d = R - |u| = v^2/(R + |u|) \leq 1/2$; then $|\cos u| = |\cos d| \geq 1/2$, while $e^{|v|} \leq e < 3$. The contour integral, divided by $2\pi i$, has absolute value at most $8/R$. Taking residues at zero and at $t_j = (j + \frac{1}{2})\pi$ gives

$$F'(0) = \sum_{|t_j| < R} \frac{(-1)^j F(t_j)}{t_j^2} + O(T^{-1}). \quad (2.3)$$

Applying the same formula to $F(z) = \sin z$ gives

$$\sum_{|t_j| < R} \frac{1}{t_j^2} = 1 + O(T^{-1}).$$

Together with $|F(t_j)| \leq 1$, equation (2.1) follows. Isolating the term $t_0 = \pi/2$ in (2.3) gives

$$1 - \delta \leq \frac{4}{\pi^2} F(\pi/2) + 1 - \frac{4}{\pi^2} + O(T^{-1}),$$

which proves (2.2). □

We apply this formula to interpolants with bounded nodal data. We first state the local estimates in the form used here, with the normalization:

$$S_n = \sum_k \frac{1}{|P'_n(x_{k,n})|}, \quad \alpha_n = \frac{1}{n} \log S_n, \quad U_n(z) = \frac{1}{n} \log \frac{1}{|P_n(z)|}.$$

Proposition 4 (Tao's local estimates). *Let $I \Subset (-1, 1)$ be a fixed nondegenerate closed interval, and suppose that $\sup_I \lambda_n \leq n$ for all sufficiently large n . This is a specialization of the polynomial upper bound in [10, equation (4.1)]. Define*

$$\rho_n(x) = -\frac{1}{\pi^2} \int_{\mathbb{R} \setminus I} \frac{U_n(y) - \alpha_n}{(y - x)^2} dy.$$

For each fixed compact interval $I' \Subset \text{int } I$, the following hold uniformly for all sufficiently large n :

1. The densities are bounded above and below by positive constants and have uniformly bounded Lipschitz norms:

$$\rho_n(x) \asymp_{I'} 1, \quad |\rho_n(x) - \rho_n(x_0)| \ll_{I'} |x - x_0| \quad (x, x_0 \in I').$$

This is the form of [10, Theorem 4.1(ii), equation (4.6)] needed here.

2. For $x \in I'$ and $\eta \geq 1/n$,

$$U_n(x + i\eta) = \alpha_n - \pi\eta\rho_n(x) + O_{I'}\left(\frac{\log n}{n} + \eta^3\right). \quad (2.4)$$

Compare [10, equation (4.7)].

3. For globally Lipschitz functions h_n supported in I' , with uniformly bounded supremum norms and Lipschitz constants,

$$\frac{1}{n} \sum_k h_n(x_{k,n}) = \int h_n(x) \rho_n(x) dx + O(n^{-1/4}). \quad (2.5)$$

The density is defined from I , as in [10, equation (4.15)], and is independent of the auxiliary interval I' . The quadrature estimate follows directly from the Poisson representation: averaging Poisson kernels between heights h and $2h$ gives an approximate identity with Lipschitz error $O(\sqrt{h})$, while (2.4) contributes $O(\log n/(nh) + h^2)$. Taking $h = n^{-1/2}$ gives (2.5).

Lemma 5. *Let $I \subseteq (-1, 1)$ be a fixed nondegenerate closed interval. Suppose that, for all sufficiently large n ,*

$$\lambda_n(x) \leq \Lambda_n := c_0 \log n + C \quad (x \in I), \quad (2.6)$$

where C is fixed. This bound gives the hypothesis of the proposition above; let ρ_n be its densities. For every fixed interval $J \subseteq \text{int } I$ and every real interpolant

$$q(x) = \sum_{k=1}^n a_k \ell_{k,n}(x), \quad |a_k| \leq 1,$$

uniformly for $x \in J$,

$$|q'(x)| \leq \pi n \rho_n(x) (\Lambda_n + o(1)), \quad |q''(x)| \leq C_J n^2 \log n. \quad (2.7)$$

Moreover, for every fixed D , if

$$\frac{|q'(x)|}{\pi n \rho_n(x)} \geq c_0 \log n - D, \quad (2.8)$$

then there is $y = x + O_J(n^{-1})$ such that

$$\lambda_n(y) \geq c_0 \log n - D', \quad (2.9)$$

where D' depends on I, J, C, D , but not on n, x, q . The point y lies in any fixed compact interior interval containing J in its interior, for all sufficiently large n .

Proof. Fix $I' \in \text{int } I$ with $J \in \text{int } I'$. For $\Re z \in I'$ and $\Im z = \eta \geq \log n/n$, the interpolation formula and (2.4) give

$$|q(z)| \leq \frac{S_n |P_n(z)|}{\eta} \leq \frac{1}{\eta} \exp\left(\pi n \eta \rho_n(\Re z) + O(\log n + n \eta^3)\right). \quad (2.10)$$

Fix $x \in J$, and put

$$L = n^{-1/5}, \quad H = n^{-2/5}, \quad \omega = \pi n \rho_n(x)(1 + n^{-1/10}).$$

On the upper edge of $\{|\Re z - x| \leq L, 0 \leq \Im z \leq H\}$, we have $|q(z)| \leq \Lambda_n e^{\omega H}$. ωH exceeds $\pi n H \rho_n(x)$ by $\pi \rho_n(x) n^{1/2}$, which dominates the $O(n^{2/5})$ from the Lipschitz variation of ρ_n and the remaining $O(\log n)$ terms in (2.10). On the lower edge $|q| \leq \Lambda_n$; a Chebyshev expansion on I gives $|q| \leq \Lambda_n \exp(O_I(n))$ on the vertical sides. For $f(z) = \Lambda_n^{-1} q(z) e^{i\omega(z-x)}$, the horizontal edges have $|f| \leq 1$, and the vertical edges have $|f| \leq e^B$ with $B = O_I(n)$. Set

$$\mathcal{B}(z) = 2B \left[e^{(z-x-L)/H} + e^{(-z+x-L)/H} \right].$$

Its real part is nonnegative on the horizontal edges and at least B on the vertical edges, since $\cos(v/H) \geq 1/2$ for $0 \leq v \leq H$. The maximum modulus principle gives $|f(z) e^{-\mathcal{B}(z)}| \leq 1$. At distance at least $L/2$ from the vertical sides, the resulting error factor is $\exp(O(ne^{-L/(2H)})) = 1 + O(n^{-10})$. Thus, also reflecting across the real axis,

$$|q(x + u + iv)| \leq \Lambda_n (1 + n^{-10}) e^{\omega|v|} \quad (|u| \leq L/2, |v| \leq H). \quad (2.11)$$

The rescaled function

$$F(t) = \frac{q(x + t/\omega)}{\Lambda_n (1 + n^{-10})}$$

satisfies Lemma 3 on a disk of radius $T \asymp_J n^{3/5}$. By (2.1),

$$|q'(x)| = \omega \Lambda_n (1 + n^{-10}) |F'(0)| \leq \pi n \rho_n(x) \Lambda_n (1 + n^{-1/10}) (1 + O(n^{-3/5})).$$

Since $\Lambda_n n^{-1/10} = o(1)$, this gives the first bound in (2.7); Cauchy's estimate on a disk of radius comparable to n^{-1} gives the second. Under (2.8), change the sign of q if necessary. For sufficiently large n ,

$$F'(0) \geq \frac{c_0 \log n - D}{\Lambda_n (1 + n^{-1/10}) (1 + n^{-10})} \geq 1 - \frac{C_1}{\Lambda_n}, \quad C_1 = 1 + \max\{C + D, 0\}.$$

Equation (2.2) then gives $|q(x + \pi/(2\omega))| \geq \Lambda_n - C_2$. Since $|q(y)| \leq \lambda_n(y)$ on the real axis and $\omega \asymp_J n$, this proves (2.9) with the asserted displacement. $\square$

3 Derivative jumps and a Riesz energy

At each internal interpolation node, write

$$J_k = \lambda'_n(x_{k,n}+) - \lambda'_n(x_{k,n}-).$$

At the first and last nodes, use the exterior polynomial pieces. Set

$$\nu_k = \frac{1}{S_n |P'_n(x_{k,n})|}.$$

The absolute values of all cardinal polynomials other than $\ell_{k,n}$ have a cusp at $x_{k,n}$, whereas $\ell_{k,n}$ is positive near that node. Using $\ell_{j,n}(x) = P_n(x)/((x-x_{j,n})P'_n(x_{j,n}))$ and $P_n(x_{k,n}) = 0$ to differentiate at $x_{k,n}$, we obtain

$$J_k = 2 \sum_{j \neq k} |\ell'_{j,n}(x_{k,n})| = 2 \sum_{j \neq k} \frac{\nu_j}{\nu_k |x_{k,n} - x_{j,n}|}. \quad (3.1)$$

Lemma 6. *Under the hypotheses of Lemma 5, choose a nonzero nonnegative smooth function f compactly supported in $\text{int } I$. Then*

$$\sum_k \frac{f(x_{k,n})^2}{n \rho_n(x_{k,n})} \frac{J_k}{2\pi n \rho_n(x_{k,n})} \geq c_0 \log n \int f(x)^2 dx - O(1). \quad (3.2)$$

Terms with $f(x_{k,n}) = 0$ are taken to be zero; ρ_n is only needed on a fixed interior neighborhood of the support of f .

Proof. Put $a_k = f(x_{k,n})/\rho_n(x_{k,n})$, with $a_k = 0$ outside the support of f , and define

$$\mu_n = \frac{1}{n} \sum_k a_k \delta_{x_{k,n}}, \quad K_\varepsilon(t) = \frac{1}{\sqrt{t^2 + \varepsilon^2}}.$$

The kernel K_ε is positive definite, since

$$K_\varepsilon(t) = \frac{1}{\sqrt{\pi}} \int_0^\infty s^{-1/2} e^{-s\varepsilon^2} e^{-st^2} ds$$

is a positive mixture of Gaussian kernels. Write

$$\mathcal{E}_\varepsilon(\sigma, \tau) = \iint K_\varepsilon(x - y) d\sigma(x) d\tau(y)$$

for its bilinear energy. Positivity applied to $\mu_n - f(x) dx$ gives

$$\mathcal{E}_\varepsilon(\mu_n, \mu_n) \geq 2\mathcal{E}_\varepsilon(\mu_n, f dx) - \mathcal{E}_\varepsilon(f dx, f dx). \quad (3.3)$$

Take $\varepsilon = 1/n$. Subtracting $f(x)$ in the integral over $|t| \leq 1$, and using the Lipschitz bound for f , gives the uniform estimate

$$(K_{1/n} * f)(x) = 2 \log n f(x) + O_f(1). \quad (3.4)$$

The mass of μ_n is bounded. Also, by (2.5),

$$\int f d\mu_n = \frac{1}{n} \sum_k \frac{f(x_{k,n})^2}{\rho_n(x_{k,n})} = \int f^2 + O(n^{-1/10}). \quad (3.5)$$

Again by (3.4),

$$\mathcal{E}_{1/n}(f dx, f dx) = 2 \log n \int f^2 + O(1).$$

Together with (3.3)–(3.5), this yields

$$\mathcal{E}_{1/n}(\mu_n, \mu_n) \geq 2 \log n \int f^2 - O(1). \quad (3.6)$$

The diagonal part of this energy is $n^{-1} \sum_k a_k^2 = O(1)$. Since $K_{1/n}(t) \leq 1/|t|$ for $t \neq 0$,

$$\frac{1}{n^2} \sum_{k \neq j} \frac{a_k a_j}{|x_{k,n} - x_{j,n}|} \geq 2 \log n \int f^2 - O(1). \quad (3.7)$$

Finally,

$$a_k^2 \frac{\nu_j}{\nu_k} + a_j^2 \frac{\nu_k}{\nu_j} \geq 2a_k a_j.$$

By (3.1), the left side of (3.2) is

$$\frac{1}{\pi n^2} \sum_{k \neq j} \frac{a_k^2 \nu_j}{\nu_k |x_{k,n} - x_{j,n}|}.$$

Pairing the terms indexed by (k, j) and (j, k) , and then applying (3.7), bounds this expression below by

$$\frac{1}{\pi n^2} \sum_{k \neq j} \frac{a_k a_j}{|x_{k,n} - x_{j,n}|} \geq \frac{2}{\pi} \log n \int f^2 - O(1).$$

This proves (3.2). □

4 A dense set of points with bounded additive loss

Proposition 7. *Assume the local upper bound (2.6) on a fixed nondegenerate interval $I \Subset (-1, 1)$. There are a constant D and a measurable set $G \subset \text{int } I$, with $|G| > 0$, such that every $x \in G$ satisfies*

$$\lambda_n(x) \geq c_0 \log n - D$$

for infinitely many n .

Proof. Choose compact intervals $J_0 \Subset J_1 \Subset \text{int } I$, and choose f as in Lemma 6 with $\text{supp } f \subset J_0$. All constants below are independent of n : after I, C, J_0, J_1, f are fixed, we choose D , then $\kappa, D_1, d, \kappa_1, d_1$, and finally b, D_2 . Let

$$w_k = \frac{f(x_{k,n})^2}{n \rho_n(x_{k,n})}, \quad T_k = \frac{J_k}{2\pi n \rho_n(x_{k,n})}, \quad W = \int f^2 > 0.$$

By (2.5),

$$\sum_k w_k = W + O(n^{-1/10}), \quad 0 \leq w_k \leq C_f/n. \quad (4.1)$$

On each interval between successive nodes, λ_n equals an interpolant with nodal data in $\{-1, 1\}$. Both one-sided derivatives at a node therefore satisfy (2.7), so

$$0 \leq T_k \leq \Lambda_n + o(1) \quad (4.2)$$

uniformly on the support of f . Lemma 6, (4.1), and (4.2) show that

$$\sum_k w_k (\Lambda_n + 1 - T_k) = O(1), \quad (4.3)$$

with nonnegative summands for all sufficiently large n . On a node with $T_k < c_0 \log n - D$, the factor in (4.3) is greater than $C + 1 + D$. Hence, by choosing D sufficiently large, such nodes carry

at most $W/2$ of the weights. Since (4.1) has total weight $W + o(1)$, for all sufficiently large n the nodes in the support of f satisfying

$$T_k \geq c_0 \log n - D \quad (4.4)$$

carry at least $W/4$ of the weights. In particular, there are at least κn such nodes, with $\kappa > 0$ independent of n . Discarding the first and last nodes does not change this conclusion after reducing κ .

At a node satisfying (4.4), at least one adjacent polynomial piece q has

$$|q'(x_{k,n})| \geq \pi n \rho_n(x_{k,n})(c_0 \log n - D).$$

Lemma 5 gives a point

$$y_k = x_{k,n} + O(n^{-1}), \quad \lambda_n(y_k) \geq c_0 \log n - D_1. \quad (4.5)$$

The same polynomial piece q has value 1 at both endpoints of the corresponding gap. Rolle's theorem and (2.7) imply that this gap has length at least d/n for some fixed $d > 0$ when the gap is contained in J_1 . If it is not contained in J_1 , its length is bounded below by $\text{dist}(J_0, \mathbb{R} \setminus J_1) > 0$.

An interval of length less than d/n contains at most two such nodes, so every interval of length $O(n^{-1})$ contains only a bounded number of the nodes in (4.4). By (4.5), the same holds, with multiplicities, for the points y_k . Hence one can select at least $\kappa_1 n$ of them, separated by d_1/n , for fixed $\kappa_1, d_1 > 0$.

Equation (2.7), applied to every polynomial piece, also shows that λ_n is $C_I n \log n$ -Lipschitz on a fixed compact interior interval containing these points. Choose a fixed sufficiently small $b > 0$, and let A_n be the union of the intervals of radius

$$r_n = \frac{b}{n \log n}$$

centered at the selected points. For all sufficiently large n , these intervals are disjoint, lie in a fixed compact interior interval, and satisfy

$$\lambda_n(x) \geq c_0 \log n - D_2 \quad (x \in A_n), \quad \frac{c}{\log n} \leq |A_n| \leq \frac{C_2}{\log n}. \quad (4.6)$$

The separation of the centers in the m -th row implies, for every interval B ,

$$|A_m \cap B| \leq C_3 \left(\frac{|B|}{\log m} + \frac{1}{m \log m} \right).$$

Summing this inequality over the at most n components of A_n gives, for $m > n$,

$$|A_n \cap A_m| \leq C_3 \left(\frac{|A_n|}{\log m} + \frac{n}{m \log m} \right). \quad (4.7)$$

Put $B_j = A_{2^j}$. Equations (4.6) and (4.7) give

$$|B_j| \asymp j^{-1}, \quad |B_j \cap B_k| \leq C_4 \left(\frac{1}{jk} + \frac{2^{j-k}}{k} \right) \quad (j < k). \quad (4.8)$$

For every fixed j_0 , the sum of the measures of B_j , $j_0 \leq j \leq N$, grows at least as a positive constant times $\log N$. By (4.8), the integral of the square of the sum of their indicators is $O((\log N)^2)$. Cauchy-Schwarz consequently gives a positive lower bound, independent of j_0 , for

$$\left| \bigcup_{j \geq j_0} B_j \right|.$$

Continuity of measure along these decreasing tail unions shows that

$$|\limsup_{j \rightarrow \infty} B_j| > 0. \quad (4.9)$$

Together with (4.6), this proves the proposition. $\square$

Proof of Theorem 1(i). Fix a nondegenerate compact interval $K_0 \subset (-1, 1)$, and suppose that it contains no point satisfying the assertion. Then

$$\lambda_n(x) - c_0 \log n \longrightarrow -\infty \quad \text{for every } x \in K_0. \quad (4.10)$$

The closed sets

$$F_M = \{x \in K_0 : \lambda_n(x) \leq c_0 \log n + M \text{ for every } n \geq 2\}, \quad M = 1, 2, \dots,$$

cover K_0 , by (4.10). Baire's theorem supplies a nondegenerate closed subinterval $I \subset K_0$ on which (2.6) holds with a fixed C . Proposition 7 contradicts (4.10). Thus every such K_0 contains a good point, proving density. Increasing the final additive constant by one gives the strict inequality in the statement. $\square$

5 Positive Cauchy transforms

Suppress the row index and put

$$S = \sum_k \frac{1}{|P'(x_k)|}, \quad \nu_k = \frac{1}{S|P'(x_k)|}, \quad \epsilon_k = \operatorname{sgn} P'(x_k).$$

Define

$$m(z) = \sum_k \frac{\nu_k}{z - x_k}, \quad g(z) = \sum_k \frac{\epsilon_k \nu_k}{z - x_k} = \frac{1}{SP(z)}. \quad (5.1)$$

The last identity is the partial fraction formula for $1/P$, whose expansion at infinity gives

$$\sum_k \epsilon_k \nu_k x_k^j = 0 \quad (0 \leq j \leq n-2). \quad (5.2)$$

For $n \geq 2$, both $m + g$ and $m - g$ are Cauchy transforms of positive probability measures: their masses at x_k are $(1 \pm \epsilon_k) \nu_k$.

For real x away from the nodes, write

$$H(x) = \sum_k \frac{\nu_k}{|x - x_k|}, \quad \gamma_s(x) = -\Im m(x + is), \quad p_s(t) = \frac{s}{\pi(t^2 + s^2)}.$$

Then

$$\lambda_n(x) = \frac{H(x)}{|g(x)|}, \quad H(x) = \frac{2}{\pi} \int_0^\infty \gamma_s(x) \frac{ds}{s}, \quad (5.3)$$

and the Poisson semigroup gives

$$\gamma_s = p_{s-h} * \gamma_h \quad (s > h). \quad (5.4)$$

Lemma 8. For every row with $n \geq 2$, every $0 < h \leq 1$, and every $x \in [-1, 1]$,

$$\frac{|g(x + ih)|}{\gamma_h(x)} \leq \frac{4 + h^2}{h^2 \sinh((n-1)h/2)}. \quad (5.5)$$

Consequently, if $0 < a < 1$ and $h = n^{-a}$, then

$$\sup_{x \in [-1, 1]} \frac{|g(x + ih)|}{\gamma_h(x)} \rightarrow 0. \quad (5.6)$$

Proof. Set $z = x + ih$ and $p = T_{n-1}$, the Chebyshev polynomial of the first kind. The polynomial $(p(t) - p(z))/(t - z)$ has degree at most $n - 2$, so (5.2) gives

$$p(z)g(z) = \sum_k \frac{\epsilon_k \nu_k p(x_k)}{z - x_k}, \quad |p(z)g(z)| \leq \frac{1}{h},$$

using $|p(x_k)| \leq 1$ and $\sum_k \nu_k = 1$. Write $z = \cos(u + iv)$, with $u, v \in \mathbb{R}$. Then $h \leq \sinh |v|$, so $|v| \geq \operatorname{arsinh} h \geq h/2$. The identity $T_{n-1}(\cos w) = \cos((n-1)w)$ yields

$$|p(z)|^2 = \cos^2((n-1)u) + \sinh^2((n-1)v) \geq \sinh^2((n-1)h/2).$$

Finally, since the nodes and x lie in $[-1, 1]$,

$$\frac{h}{4 + h^2} \leq \gamma_h(x) = \sum_k \frac{\nu_k h}{(x - x_k)^2 + h^2}.$$

Combining these bounds proves (5.5). For $h = n^{-a}$, its right-hand side tends to zero: the hyperbolic sine grows exponentially in n^{1-a} , while h^{-2} grows polynomially. This proves (5.6). $\square$

Lemma 9. Let F be the Cauchy transform of a positive probability measure with finite support. For every bounded real interval S_0 , every $x \in \mathbb{R}$, and every $h > 0$,

$$\int_{\mathbb{R}} p_h(x - y) \mathbf{1}_{\{F(y) \in S_0\}} dy = \frac{1}{\pi} \int_{S_0} \frac{-\Im F(x + ih)}{(t - \Re F(x + ih))^2 + (\Im F(x + ih))^2} dt. \quad (5.7)$$

Proof. The function F maps the upper half-plane into the lower half-plane. Compose harmonic measure of S_0 there with F . The resulting bounded harmonic function on the upper half-plane has the indicated boundary values almost everywhere. The exceptional boundary points are poles of F and preimages of the endpoints of S_0 , a finite set. Its Poisson representation is (5.7). $\square$

6 The almost-everywhere bound

Proof of Theorem 1(ii). The failure set is the union, over rational $0 < c < c_0$ and integers $N \geq 2$, of the measurable sets

$$\{y \in (-1, 1) : \lambda_n(y) \leq c \log n \text{ for every } n \geq N\}.$$

If the assertion fails, one of these sets E has positive measure. For its fixed c, N , we therefore have

$$\lambda_n(y) \leq c \log n \quad (y \in E, n \geq N). \quad (6.1)$$

Choose constants

$$K > 1, \quad 0 < b < a < 1, \quad \frac{2b}{\pi cK} > 1,$$

which is possible because $c < 2/\pi$. Set

$$h = n^{-a}, \quad \eta = n^{-b}, \quad H^\eta(x) = \frac{2}{\pi} \int_\eta^1 \gamma_s(x) \frac{ds}{s}.$$

By (5.3), $H^\eta \leq H$. For $s > 0$ and real x, y, t , the triangle inequality gives

$$|y + is - t| \leq |x + is - t| + |x - y| \leq \left(1 + \frac{|x - y|}{s}\right) |x + is - t|.$$

Taking reciprocal squares and summing against the positive masses yields $\gamma_s(x) \leq (1 + |x - y|/s)^2 \gamma_s(y)$. Integrating over $\eta \leq s \leq 1$, we obtain

$$H^\eta(x) \leq \left(1 + \frac{|x - y|}{\eta}\right)^2 H^\eta(y). \quad (6.2)$$

Fix $x \in E$, and write

$$u = \Re m(x + ih), \quad \gamma = \gamma_h(x), \quad S_x = [u - K\gamma, u + K\gamma].$$

By (5.6) and (5.7), each of the two events

$$m(y) + g(y) \in S_x, \quad m(y) - g(y) \in S_x$$

has probability

$$p_K + o(1), \quad p_K = \frac{2}{\pi} \arctan K > \frac{1}{2},$$

under the Poisson probability $p_h(x - y) dy$. The error is uniform in $x \in E$: for $F = m \pm g$, (5.6) gives $\Re F(x + ih) = u + o(\gamma)$ and $-\Im F(x + ih) = \gamma(1 + o(1))$. Substituting $t = u + \gamma s$ in (5.7) therefore gives $\pi^{-1} \int_{-K}^K (1 + s^2)^{-1} ds + o(1) = p_K + o(1)$. The two events have intersection probability at least

$$q_K + o(1), \quad q_K = 2p_K - 1 > 0. \quad (6.3)$$

Choose R so that the Poisson mass of $|y - x| > Rh$ is less than $q_K/8$, and let

$$E_n = \{x \in E : (p_h * \mathbf{1}_{\mathbb{R} \setminus E})(x) < q_K/8\}.$$

The approximate identity theorem gives $|E \setminus E_n| \rightarrow 0$. For all sufficiently large n , (6.3) is greater than $q_K/2$. For $x \in E_n$, the two excluded sets $\mathbb{R} \setminus E$ and $\{|y - x| > Rh\}$ have total Poisson mass less than $q_K/4$. Hence there exists $y \in E$ with $|y - x| \leq Rh$ and both events true. We may take y away from the nodes. Since $m(y) \pm g(y)$ both lie in an interval of length $2K\gamma_h(x)$,

$$|g(y)| \leq K\gamma_h(x).$$

Equations (5.3) and (6.1) give

$$H^\eta(y) \leq H(y) = \lambda_n(y) |g(y)| \leq cK(\log n) \gamma_h(x).$$

Together with (6.2), this yields

$$H^\eta(x) \leq (1 + Rh/\eta)^2 H^\eta(y) \leq cK(1 + Rh/\eta)^2 (\log n) \gamma_h(x) \quad (x \in E_n). \quad (6.4)$$

Let T_n be convolution with the symmetric probability kernel

$$k_n(t) = \frac{1}{b \log n} \int_{\eta}^1 p_{s-h}(t) \frac{ds}{s}.$$

Its integral is one, since $\int_{\eta}^1 ds/s = b \log n$. Equation (5.4) then shows that

$$H^{\eta} = \frac{2b}{\pi} (\log n) T_n \gamma_h.$$

Thus (6.4) becomes

$$\frac{T_n \gamma_h(x)}{\gamma_h(x)} \leq \frac{1}{A_n} \quad (x \in E_n), \quad A_n = \frac{2b}{\pi c K} (1 + Rh/\eta)^{-2} \longrightarrow A > 1. \quad (6.5)$$

Integrate (6.5) over E_n , discard the contribution from $y \notin E_n$, and symmetrize. Positivity of γ_h , symmetry of k_n , and $(r + r^{-1})/2 \geq 1$ give

$$\frac{|E_n|}{A_n} \geq \iint_{E_n \times E_n} k_n(x-y) \frac{\gamma_h(y)}{\gamma_h(x)} dy dx \geq \iint_{E_n \times E_n} k_n(x-y) dy dx. \quad (6.6)$$

The kernels k_n form an approximate identity. In fact, for every fixed $\delta > 0$,

$$\int_{|t|>\delta} k_n(t) dt \leq \frac{C}{b \log n} \int_{\eta}^1 \min(1, s/\delta) \frac{ds}{s} \longrightarrow 0.$$

It follows that

$$\iint_{E \times E} k_n(x-y) dy dx \longrightarrow |E|.$$

Replacing E by E_n changes this integral by at most $2|E \setminus E_n|$. Taking limits in (6.6) therefore yields

$$\frac{|E|}{A} \geq |E|,$$

contrary to $|E| > 0$ and $A > 1$. This proves the assertion. $\square$

7 Failure of uniform additive constants

We prove Theorem 2. Smooth on $I = [-1, 1]$ will mean extendible to a smooth function on a neighborhood of I .

7.1 The compact set

For $I = [-1, 1]$, write

$$Lf(x) = \int_I \frac{f(x) - f(y)}{|x - y|} dy.$$

In particular,

$$\|Lf\|_{\infty} \leq 2\|f'\|_{\infty} \quad (f \in C^1(I)). \quad (7.1)$$

Fix $A > 1$, and put

$$\rho = e^{-1024(A+40)}, \quad \delta_k = \rho 2^{-k}, \quad m_k = \lceil e^{1/\delta_k} \rceil.$$

Starting with I , replace every retained interval of length ℓ_{k-1} at stage k by m_k equally long closed intervals, retaining both endpoints of the parent. The children and intervening open gaps have respective lengths

$$\ell_k = \frac{(1 - \delta_k)\ell_{k-1}}{m_k}, \quad b_k = \frac{\delta_k\ell_{k-1}}{m_k - 1}, \quad \ell_0 = 2.$$

Let F be the intersection of the retained sets and $U = I \setminus F$. Thus F is compact, contains both endpoints, and has empty interior, since $\ell_k \rightarrow 0$. The set U is a dense open union of disjoint gaps, and

$$|U| \leq 2 \sum_k \delta_k = 2\rho. \quad (7.2)$$

We need the following two properties:

$$|F \cap (x - r, x + r)| \geq \kappa r \quad (x \in F, \ 0 < r \leq 1), \quad \kappa = 1/32, \quad (7.3)$$

and

$$\int_U \frac{dy}{|x - y|} = \infty \quad (x \in F). \quad (7.4)$$

For (7.3), choose the first k for which $\ell_k \leq r/16$, and let J be the stage $k - 1$ parent containing x , with $|J| > r/16$. At least half the length of any retained interval belongs to F , since every subsequent product of retained proportions is at least $1 - \rho > 1/2$. If $|J| \leq r$, then $J \subset [x - r, x + r]$, proving (7.3). Otherwise $J \cap [x - r, x + r]$ has length $s \geq r$. Write $a = \ell_k$, $b = b_k$, $\delta = \delta_k$. Then

$$b \leq a, \quad b/(a + b) \leq 2\delta.$$

The gaps intersecting an interval of length s have total length at most $2\delta s + 2b$. The two partially intersected children have total length at most $2a$. The full children inside this interval therefore have total length at least

$$(1 - 2\delta)s - 2(a + b) \geq r/2.$$

Their intersections with F have total length at least $r/4$, as required.

For (7.4), in the stage $k - 1$ parent containing x , one side of its child contains at least $(m_k - 1)/2$ gaps. Use the first $q = \lfloor m_k/8 \rfloor$ on that side. The far endpoint of the j -th such gap is at distance at most $4j\ell_{k-1}/m_k$ from x , and its length is at least $\delta_k\ell_{k-1}/m_k$. These gaps contribute at least

$$\frac{\delta_k}{4} \sum_{j=1}^q \frac{1}{j} \geq \frac{\delta_k}{4} (\log m_k - \log 16) \geq \frac{1}{8}.$$

The gaps used at different stages are disjoint. Summation proves (7.4).

7.2 An explicit logarithmic integral estimate

On each gap (b, c) , define

$$r(x) = \frac{(x - b)(c - x)}{c - b},$$

and put $r = 0$ on F . If $D(x) = \text{dist}(x, F)$, then

$$r(x) \leq D(x) \leq 2r(x) \quad (x \in U). \quad (7.5)$$

The function r is globally 1-Lipschitz: its derivative has absolute value at most one on each gap, and points in distinct gaps can be compared through a point of F between them.

Put

$$B = e^8, \quad \alpha = 1/256, \quad p(t) = [\log(B/t)]^{-\alpha}, \quad u(x) = p(r(x)) \quad (x \in U), \quad u = 0 \quad (x \in F). \quad (7.6)$$

Then u is continuous on I , bounded, positive and smooth in each gap, and vanishes on F . In particular, $Lu(x)$ is an absolutely convergent integral for every $x \in U$.

We prove the quantitative bound

$$\frac{Lu(x)}{u(x)} \geq \frac{1}{512} \log \frac{1}{4r(x)} - 32 \quad \text{if } \log(B/r(x)) \geq 16. \quad (7.7)$$

All estimates below apply uniformly up to the endpoints of I .

Fix such an x , and write $s = r(x)$, $T = \log(B/s)$, $a_0 = 4s$. For $t \in [0, 2]$, let

$$M_I(t) = |I \cap (x - t, x + t)|, \quad M_U(t) = |U \cap (x - t, x + t)|.$$

Choose $z \in F$ with $|x - z| = D(x) \leq 2s$. If $4s \leq t \leq 2$, the ball of radius $t/2$ about z , intersected with I , is contained in the ball of radius t about x . By (7.3), the latter contains at least $\kappa t/2$ of F . Since $M_I(t) \leq 2t$,

$$M_U(t) \leq (1 - \eta)M_I(t), \quad \eta = \kappa/4 = 1/128 \quad (4s \leq t \leq 2). \quad (7.8)$$

First consider $|y - x| < 4s$. For $|y - x| < s/2$, Lipschitz continuity of r gives $r(y) \in [s/2, 3s/2]$. On this range $p'(r(y)) \leq 4p(s)/s$, by the derivative formula for p and $T \geq 16$. The integral of the absolute difference quotient over this part is thus at most $4p(s)$. On $s/2 \leq |y - x| < 4s$, we have $r(y) \leq 5s$ and

$$p(5s)/p(s) = \left(\frac{T}{T - \log 5} \right)^\alpha \leq 2.$$

The absolute difference is at most $3p(s)$, so this part contributes at most $6 \log 8 p(s) < 18p(s)$. In particular, the whole near contribution to $Lu(x)$ is at least $-24p(s)$.

For the far contribution, set $l_- = 1 + x$, $l_+ = 1 - x$, and

$$H = \sum_{\nu \in \{-, +\}} \max\{\log(l_\nu/(4s)), 0\}.$$

The positive part is exactly $p(s)H$. For the negative part, $r(y) \leq s + |x - y| \leq (5/4)|x - y|$, so it is bounded by

$$\int_{a_0}^2 f(t) dM_U(t), \quad f(t) = \frac{p(5t/4)}{t}.$$

The function f is decreasing. Stieltjes integration by parts, dropping the nonpositive lower-end boundary term and using (7.8), gives

$$\int_{a_0}^2 f(t) dM_U(t) \leq (1 - \eta) \left[f(a_0)M_I(a_0) + \int_{a_0}^2 f(t) dM_I(t) \right]. \quad (7.9)$$

The first term on the right, including its factor, is at most $4p(s)$.

Since $dM_I(t) = 1_{\{t < l_-\}} dt + 1_{\{t < l_+\}} dt$, the remaining term splits into at most two integrals. For a side with $l_\nu > 4s$, put

$$h = \log(l_\nu/(4s)), \quad T_0 = T - \log 5.$$

Then $0 \leq h < T_0$, since $T_0 - h = \log(4B/(5l_\nu)) > 0$, and

$$\int_{4s}^{l_\nu} \frac{p(5t/4)}{t} dt = \int_0^h (T_0 - z)^{-\alpha} dz \leq \frac{T_0^{-\alpha}}{1 - \alpha} h. \quad (7.10)$$

The last inequality follows from $(1 - z)^{1-\alpha} \geq 1 - z$ for $0 \leq z \leq 1$.

Finally, concavity of the power α , $\log 5 < 2$, and $T \geq 16$ give

$$\begin{aligned} \frac{1 - \eta T_0^{-\alpha}}{1 - \alpha p(s)} &= \frac{254}{255} \left(\frac{T}{T - \log 5} \right)^{1/256} \\ &\leq \frac{254}{255} (8/7)^{1/256} \leq \frac{254}{255} \left(1 + \frac{1}{1792} \right) \leq 1 - \frac{1}{512}. \end{aligned} \quad (7.11)$$

Thus the negative far contribution is at most $4p(s) + (1 - 1/512)p(s)H$. Including the near part gives $Lu(x)/u(x) \geq H/512 - 28$. At least one of l_-, l_+ is at least one, so $H \geq \log(1/(4s))$. This proves (7.7).

Every gap has length at most $|U| \leq 2\rho$, so $r(x) \leq \rho/2$. Our choice of ρ ensures $T \geq 16$, and (7.7) yields

$$\frac{Lu(x)}{u(x)} \geq 2(A + 40) - \frac{\log 2}{512} - 32 > A + 2 \quad (x \in U). \quad (7.12)$$

It also shows that $Lu(x)/u(x) \rightarrow +\infty$ as x approaches either endpoint of any fixed gap.

7.3 Smooth positive functions on the whole interval

Let K_N be the union of the closed middle halves of all gaps removed in the first N stages. This is a compact subset of U . For any smooth $0 \leq \chi \leq 1$, vanishing near F and equal to one on K_N , put

$$A_\chi(x) = \int_U \frac{\chi(y)}{|x - y|} dy \quad (x \in F).$$

The estimate proving (7.4), restricted to these middle halves, gives at least $1/16$ from each of the first N stages. Consequently,

$$A_\chi(x) \geq N/16 \quad (x \in F). \quad (7.13)$$

Let $\epsilon_j = 1/j$, $j \geq 2$, and put

$$d_j = \exp(8 - (2j)^{256}), \quad C_j = 2(2j)^{256}.$$

Since $r \leq D$ and $p(d_j) = 1/(2j)$, one has $u(y) \leq \epsilon_j/2$ when $D(y) \leq d_j$. Using this for $|x - y| < d_j$, and $u \leq 1$ on the remainder, gives

$$\int_U \frac{\chi(y)u(y)}{|x - y|} dy \leq \frac{\epsilon_j}{2} A_\chi(x) + C_j \quad (x \in F). \quad (7.14)$$

Here the far integral is at most $2 \log(2/d_j) \leq C_j$.

Set

$$k_j = \lceil 32(A + 1) \rceil + 64j(2j)^{256} + j.$$

Choose a smooth function $f : \mathbb{R} \rightarrow [0, 1]$ with $\{f > 0\} = U$, and a nondecreasing smooth $q_0 : \mathbb{R} \rightarrow [0, 1]$, equal to zero on $(-\infty, 1]$ and one on $[2, \infty)$. By compactness, choose $0 < \tau_j \leq 1/j$ such that $2\tau_j \leq f$ on K_{k_j} , and define

$$\chi_j(x) = q_0(f(x)/\tau_j).$$

Thus χ_j is smooth, vanishes on a neighborhood of F , and equals one on K_{k_j} . Since $\tau_j \rightarrow 0$, at each fixed point of U it is eventually identically one on a neighborhood of that point. Equation (7.13) gives

$$\frac{\epsilon_j}{2} A_{\chi_j}(x) - C_j \geq \frac{k_j}{32j} - C_j \geq (A+1)\epsilon_j \quad (x \in F).$$

Define

$$v_j = \chi_j u + (1 - \chi_j)\epsilon_j. \quad (7.15)$$

This function is strictly positive and smooth on a neighborhood of I : the function u is smooth and positive in each gap, and $v_j = \epsilon_j$ near F .

At $x \in F$, (7.14) gives $Lv_j(x) \geq (A+1)v_j(x)$. At every fixed $x \in U$, eventually $v_j = u$ on a neighborhood of x . Also $v_j \rightarrow u$ in $L^1(I)$, by bounded convergence. The integrals outside that neighborhood then converge, so

$$\frac{Lv_j(x)}{v_j(x)} \rightarrow \frac{Lu(x)}{u(x)} > A+2 \quad (x \in U). \quad (7.16)$$

We have therefore constructed positive smooth functions such that

$$\frac{Lv_j(x)}{v_j(x)} \geq A+1 \quad \text{eventually for every fixed } x \in I. \quad (7.17)$$

The eventual index is allowed to depend on x .

7.4 From amplitudes to interpolation nodes

We prove an upper estimate with an absolute additive constant.

Lemma 10. *Let $h(z) = \sum_{r=0}^d h_r z^r$ have real coefficients, no zero on the closed unit disk, and $h(0) > 0$. Set*

$$v(\cos \theta) = |h(e^{i\theta})|^{-1}.$$

In this subsection, T_k denotes the Chebyshev polynomial of the first kind, defined by $T_k(\cos \theta) = \cos(k\theta)$. For $n > d$, define the real polynomial

$$P_n(x) = \sum_{r=0}^d h_r T_{n-r}(x). \quad (7.18)$$

For all sufficiently large n , this polynomial has exactly n simple roots in $(-1, 1)$. The corresponding Lebesgue function satisfies, on any fixed $K \Subset (-1, 1)$,

$$\lambda_n(x) \leq \frac{2}{\pi} \log n + 3 - \frac{Lv(x)}{\pi v(x)} \quad (x \in K) \quad (7.19)$$

for all sufficiently large n , with the threshold allowed to depend on h, K .

Proof. The analytic logarithm of h , chosen real at zero, is real on the real diameter. Its imaginary part $\psi(\theta) = \arg h(e^{i\theta})$ thus satisfies $\psi(0) = \psi(\pi) = 0$. We have the exact identity

$$P_n(\cos \theta) = |h(e^{i\theta})| \cos(n\theta - \psi(\theta)).$$

For $n > \|\psi'\|_\infty$, the phase increases strictly from 0 to $n\pi$. The polynomial has degree exactly n , since its leading term comes from $h_0 T_n$, with $h_0 > 0$. This gives the asserted n simple roots. At their angles θ_k ,

$$|P'_n(\cos \theta_k)| = |h(e^{i\theta_k})| \frac{n - \psi'(\theta_k)}{\sin \theta_k}. \quad (7.20)$$

Put $s = (n\theta - \psi(\theta))/n$, let $\theta = \Theta_n(s)$ be the inverse, and set $X_n(s) = \cos \Theta_n(s)$. The roots correspond to $s_k = (k - \frac{1}{2})\pi/n$. Moreover $X_n \rightarrow \cos$ in $C^3([0, \pi])$. At a point $x = X_n(s)$ other than a node, put $b = |\cos(ns)|$. Formula (7.20) gives

$$\lambda_n(x) = \frac{b}{nv(x)} \sum_{k=1}^n \frac{v(X_n(s_k))|X'_n(s_k)|}{|X_n(s) - X_n(s_k)|}. \quad (7.21)$$

Subtract the simple singularity by defining

$$R_n(s, t) = \frac{v(X_n(t))|X'_n(t)|}{|X_n(s) - X_n(t)|} - \frac{v(X_n(s))}{|s - t|}.$$

For s in a fixed compact subinterval of $(0, \pi)$, this function of t has uniformly bounded variation, with at most a bounded jump at $t = s$. To verify this near s , factor $X_n(t) - X_n(s) = (t - s)D_n(s, t)$. Both D_n and X'_n are uniformly bounded away from zero there; the numerator remaining after subtraction vanishes at $t = s$, so division leaves a uniformly C^1 function on each side, multiplied by $\text{sgn}(t - s)$. Away from s , the denominator stays uniformly away from zero. Midpoint quadrature therefore has error $O(1/n)$, uniformly also when s is arbitrarily close to a quadrature point.

Changing variables $y = X_n(t)$ in the integral, and first deleting $(s - \epsilon, s + \epsilon)$, gives

$$\begin{aligned} \int_0^\pi R_n(s, t) dt &= -Lv(x) + v(x) \log \frac{1 - x^2}{X'_n(s)^2 s(\pi - s)} \\ &= -Lv(x) + v(x) [2 \log(1 - \psi'(\theta)/n) - \log(s(\pi - s))]. \end{aligned} \quad (7.22)$$

The cancellation follows because the deleted distances in the y variable are $|X'_n(s)|\epsilon + O(\epsilon^2)$ on both sides.

Write

$$\tau = \frac{ns}{\pi} + \frac{1}{2} = m + t, \quad m = \lfloor \tau \rfloor, \quad 0 < t < 1.$$

Since $(s - s_k)n/\pi = \tau - k$, midpoint quadrature in (7.21) gives explicitly

$$\lambda_n(x) = \frac{b}{\pi} \left[\sum_{k=1}^n \frac{1}{|\tau - k|} + \frac{1}{v(x)} \int_0^\pi R_n(s, t') dt' \right] + O(1/n).$$

On the compact interval under consideration, $1 \leq m \leq n - 1$ for large n , and $b = \sin(\pi t)$. Splitting the elementary harmonic sum at m gives

$$\sum_{k=1}^n \frac{1}{|\tau - k|} \leq \log(m(n - m)) + 2 + \frac{1}{t} + \frac{1}{1 - t}.$$

Also, uniformly in this compact interval,

$$\log(m(n - m)) - \log(s(\pi - s)) = 2 \log n - 2 \log \pi + O(1/n).$$

Substituting these estimates and (7.22) into (7.21), and using $b/t \leq \pi$, $b/(1-t) \leq \pi$, yields

$$\lambda_n(x) \leq b \left[\frac{2}{\pi} \log n - \frac{Lv(x)}{\pi v(x)} + \frac{2 - 2 \log \pi}{\pi} \right] + 2 + O(1/n).$$

For large n the bracket is nonnegative, uniformly on K . We may replace $b \leq 1$ by one. Since $2 + (2 - 2 \log \pi)/\pi < 3$, this proves (7.19) away from the nodes. At the nodes, $\lambda_n = 1$, and (7.19) follows for large n as well. $\square$

Every positive smooth function v on I can be approximated in $C^1(I)$ by amplitudes occurring in this lemma. Approximate $1/v$ by a strictly positive real polynomial q of degree d , and form

$$J(z) = z^d q((z + z^{-1})/2).$$

This is a real polynomial, and $|J(e^{i\theta})| = q(\cos \theta) > 0$. In its factorization, replace every factor $z - r$ with $|r| < 1$ by $1 - \bar{r}z$, retaining the leading scalar. The resulting polynomial h has no zero on the closed disk and the same modulus as J on the unit circle. Conjugate roots are reflected together, so its coefficients are real; changing its sign makes $h(0) > 0$. Thus

$$|h(e^{i\theta})| = q(\cos \theta), \quad |h(e^{i\theta})|^{-1} = q(\cos \theta)^{-1}.$$

Polynomial approximation in C^1 follows by uniformly approximating the derivative and then integrating. Inversion is continuous in C^1 on functions with a positive lower bound. Hence q^{-1} approximates v in $C^1(I)$, and (7.1) gives arbitrarily accurate uniform approximation of Lv/v .

Remark (Dependence on interval length). Let $I_\ell = [-\ell/2, \ell/2]$, where $0 < \ell < 1/4$, and choose a smooth v equal to 1 on I_ℓ and to $\delta = 1/\log(1/\ell)$ outside $[-\ell, \ell]$, with $\delta \leq v \leq 1$ and $\|v'\|_\infty = O(1/\ell)$. Then, uniformly on I_ℓ ,

$$Lv/v = 2 \log(1/\ell) + O(1).$$

After amplitude approximation with error at most one in this quotient, Lemma 10 and the phase-midpoint calculation in (7.21)–(7.22) give

$$\sup_{I_\ell} \lambda_n = \frac{2}{\pi} \log(n\ell) + O(1).$$

for sufficiently large n , with an absolute $O(1)$ term and a starting index depending on ℓ . Thus the interval constant can be at least $(2/\pi) \log(1/|I_\ell|) - O(1)$ when the array changes with ℓ .

7.5 Assembly of the array

Proof of Theorem 2. Fix $M > 0$, and carry out Subsections 7.1–7.3 with $A = \pi(M + 4)$. For each $j \geq 2$, choose an amplitude $\tilde{v}_j = |h_j|^{-1}$ as in Subsection 7.4 so close to v_j that

$$\left\| \frac{L\tilde{v}_j}{\tilde{v}_j} - \frac{Lv_j}{v_j} \right\|_\infty < 1/2. \quad (7.23)$$

Let $K_j = [-1 + 1/j, 1 - 1/j]$. Choose a strictly increasing sequence of integers N_j , each larger than $\deg h_j$ and the threshold in Lemma 10 for h_j, K_j . For $N_j \leq n < N_{j+1}$, use as row n the n roots of (7.18) with $h = h_j$. Fill the finitely many earlier rows with Chebyshev roots.

Fix $x \in (-1, 1)$. The block index j tends to infinity with n , and $x \in K_j$ for all large j . Equations (7.17) and (7.23) give $L\tilde{v}_j(x)/\tilde{v}_j(x) \geq A + 1/2$ eventually. Hence (7.19) gives, for all sufficiently large n ,

$$\lambda_n(x) \leq \frac{2}{\pi} \log n + 3 - \frac{A + 1/2}{\pi} < \frac{2}{\pi} \log n - M.$$

This proves (1.3) and the failure of an absolute additive constant.

For the final assertion, (7.16), (7.19), and (7.23) give at every fixed $x \in U$

$$\limsup_{n \rightarrow \infty} \left(\lambda_n(x) - \frac{2}{\pi} \log n \right) \leq 3 + \frac{1}{2\pi} - \frac{Lu(x)}{\pi u(x)}. \quad (7.24)$$

Choose a gap (b, c) with both endpoints in $(-1, 1)$. By (7.7), $Lu(x)/u(x) \rightarrow +\infty$ as $x \downarrow b$. Given any finite C , there is a nonempty interval $(b, b + \sigma)$ on which the right side of (7.24) is strictly less than $-C - 1$. This interval is disjoint from G_C , proving the second assertion. $\square$

8 Consequences

Corollary 11 (Positive-measure sets). *Let $E \subset (-1, 1)$ be measurable with $|E| > 0$, and let $0 < c < 2/\pi$. For all sufficiently large n , there is $x \in E$ such that $\lambda_n(x) > c \log n$.*

Proof. Otherwise $\lambda_n(y) \leq c \log n$ for every $y \in E$ and every n in an infinite set S . The argument in Section 6 applies to these rows, with all limits taken as $n \rightarrow \infty$ through S , and gives the same contradiction $|E|/A \geq |E|$ with $A > 1$. $\square$

Corollary 12 (No uniform interval constant). *For the array satisfying both assertions of Theorem 2, with any $M > 0$, there is no finite C such that*

$$\sup_{x \in I} \lambda_n(x) \geq \frac{2}{\pi} \log n - C$$

for every nondegenerate compact interval $I \subset (-1, 1)$ and all sufficiently large n , with the threshold allowed to depend on I .

Proof. Suppose that such a C exists. If G_{C+1} misses a nondegenerate compact interval $K_0 \subset (-1, 1)$, the closed sets

$$F_N = \{x \in K_0 : \lambda_n(x) \leq (2/\pi) \log n - C - 1 \text{ for all } n \geq N\}$$

cover K_0 . Baire's theorem gives an F_N containing a nondegenerate compact subinterval I . The resulting upper bound on I contradicts the assumed lower bound for large n . Thus G_{C+1} is dense, contrary to Theorem 2. $\square$

Code availability

The Lean 4 formalization is available at <https://github.com/FireflySentinel/erdos-1132>.

Declaration of generative AI and AI-assisted technologies in the writing process

GPT-6 Astra proposed the derivative-jump and positive Cauchy-transform arguments and the counterexample construction, drafted the manuscript, and generated the Lean formalization. GPT-5.6 Sol and Claude Opus 5 were used to organize earlier research material and to review the Lean code. The author checked the arguments, completed the final manuscript, and takes full responsibility for the content.

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